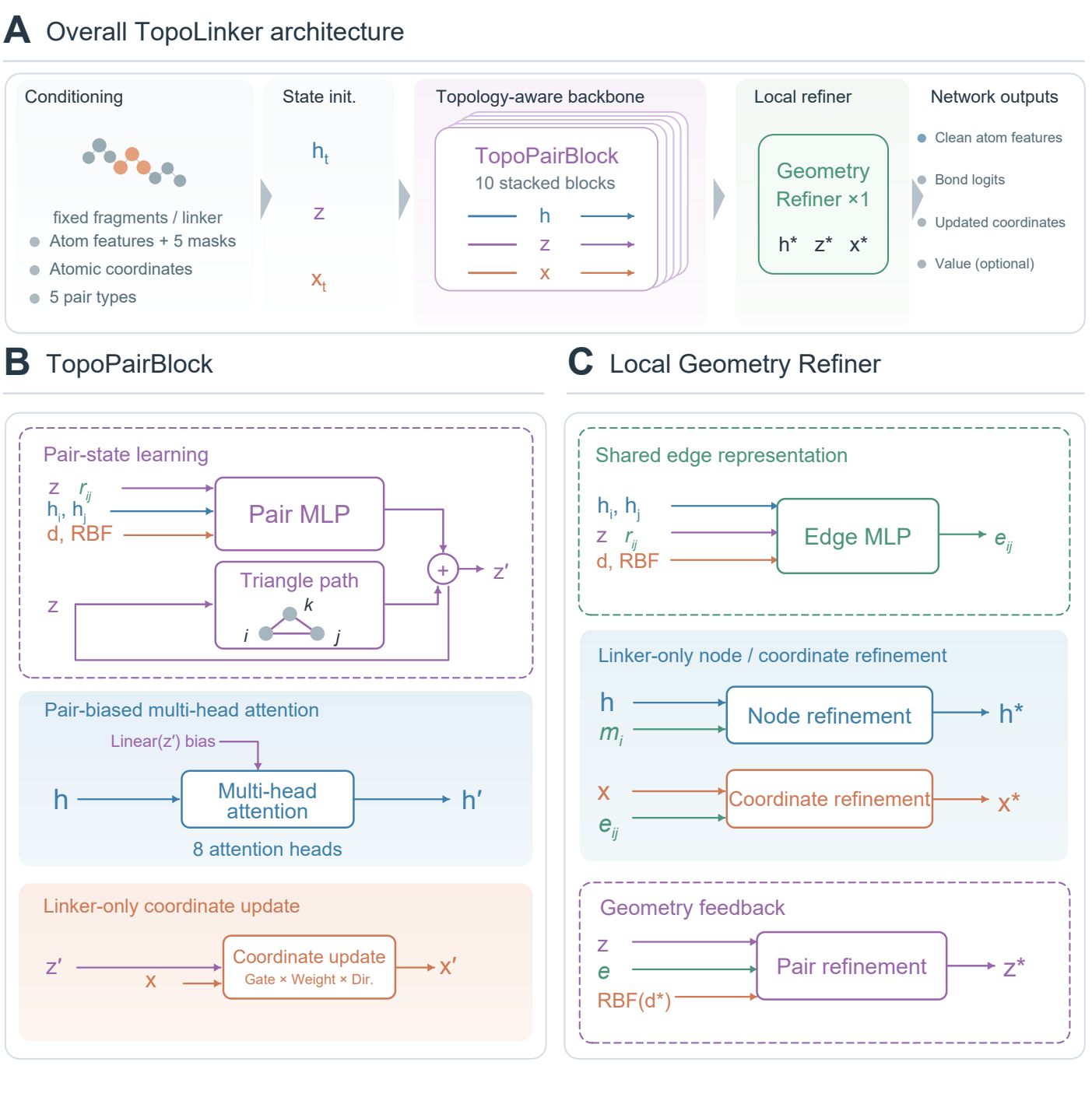

A Overall TopoLinker architecture
Conditioning
fixed fragments / linker
Atom features + 5 masks
Atomic coordinates
5 pair types
State init.
h_t
z
x_t
Topology-aware backbone
TopoPairBlock
10 stacked blocks
h
z
x
Local refiner
Geometry Refiner ×1
h* z* x*
Network outputs
Clean atom features
Bond logits
Updated coordinates
Value (optional)
B TopoPairBlock
Pair-state learning
z r_{ij}
h_i, h_j
d, RBF
Pair MLP
Triangle path
k
i
j
z
z′
Pair-biased multi-head attention
Linear(z′) bias
h
Multi-head attention
h′
8 attention heads
Linker-only coordinate update
z′
x
Coordinate update
Gate × Weight × Dir.
x′
C Local Geometry Refiner
Shared edge representation
h_i, h_j
z r_{ij}
d, RBF
Edge MLP
e_{ij}
Linker-only node / coordinate refinement
h
m_i
Node refinement
h*
x
e_{ij}
Coordinate refinement
x*
Geometry feedback
z
e
RBF(d*)
Pair refinement
z*